

#Shell Scripting

How to write and run in Vim

```
vim script.sh
```

Press **i** to enter **Insert mode**, type the script, then press **Esc** and type:

```
:wq
```

Make the script executable:

```
chmod +x script.sh
```

Run it:

```
./script.sh
```

or

```
bash script.sh
```

1. Print the message "Welcome to Shell Scripting"

```
#!/bin/bash  
echo "Welcome to Shell Scripting"
```

2. Create a variable `name`, store your name, and print it

```
#!/bin/bash  
name="Your Name"  
echo "$name"
```

3. Create three variables: `name`, `age`, and `city`, and display their values

```
#!/bin/bash  
name="Your Name"  
age=20
```

```
city="Pune"
echo "Name: $name"
echo "Age: $age"
echo "City: $city"
```

4. Store your name and course name in variables and display them on separate lines

```
#!/bin/bash
name="Your Name"
course="Linux Shell Scripting"
echo "$name"
echo "$course"
```

5. Accept the user's name and display a welcome message

```
#!/bin/bash
echo "Enter your name:"
read name
echo "Welcome, $name!"
```

6. Accept two numbers and display both numbers

```
#!/bin/bash
echo "Enter the first number:"
read num1
echo "Enter the second number:"
read num2
echo "First number: $num1"
echo "Second number: $num2"
```

7. Accept the user's name, age, and course and display all information

```
#!/bin/bash
echo "Enter your name:"
read name
echo "Enter your age:"
read age
echo "Enter your course:"
read course
echo "Name: $name"
echo "Age: $age"
echo "Course: $course"
```

8. Accept two numbers and calculate their sum

```
#!/bin/bash
echo "Enter first number:"
read num1
echo "Enter second number:"
read num2
sum=$((num1 + num2))
echo "Sum = $sum"
```

9. Accept two numbers and calculate Addition, Subtraction, Multiplication, and Division

```
#!/bin/bash
echo "Enter first number:"
read num1
echo "Enter second number:"
read num2
addition=$((num1 + num2))
subtraction=$((num1 - num2))
multiplication=$((num1 * num2))
division=$((num1 / num2))
echo "Addition = $addition"
echo "Subtraction = $subtraction"
echo "Multiplication = $multiplication"
echo "Division = $division"
```

10. Accept the length and breadth of a rectangle and calculate its area

```
#!/bin/bash
echo "Enter the length:"
read length
echo "Enter the breadth:"
read breadth
area=$((length * breadth))
echo "Area of rectangle = $area"
```

11. Accept a number and calculate its square

```
#!/bin/bash
echo "Enter a number:"
read num
square=$((num * num))
echo "Square of $num = $square"
```

12. Accept Principal, Rate, and Time and calculate Simple Interest

```
#!/bin/bash
echo "Enter Principal amount:"
read principal
echo "Enter Rate:"
read rate
echo "Enter Time:"
read time
simple_interest=$((principal * rate * time / 100))
echo "Simple Interest = $simple_interest"
```

13. Check whether a number is positive, negative, or zero

```
#!/bin/bash
echo "Enter a number:"
read num
if [ "$num" -gt 0 ]
then
    echo "Positive number"
elif [ "$num" -lt 0 ]
then
    echo "Negative number"
else
    echo "Zero"
fi
```

14. Check whether a number is even or odd

```
#!/bin/bash
echo "Enter a number:"
read num
if [ $((num % 2)) -eq 0 ]
then
    echo "Even number"
else
    echo "Odd number"
fi
```

15. Accept two numbers and display the greater number

```
#!/bin/bash
echo "Enter first number:"
read num1
echo "Enter second number:"
read num2
```

```
if [ "$num1" -gt "$num2" ]
then
    echo "$num1 is greater"
else
    echo "$num2 is greater"
fi
```

16. Accept two numbers and check whether they are equal or different

```
#!/bin/bash
echo "Enter first number:"
read num1
echo "Enter second number:"
read num2
if [ "$num1" -eq "$num2" ]
then
    echo "Both numbers are equal"
else
    echo "Numbers are different"
fi
```

17. Check whether a student has Pass or Fail marks

```
#!/bin/bash
echo "Enter student's marks:"
read marks
if [ "$marks" -ge 40 ]
then
    echo "Pass"
else
    echo "Fail"
fi
```

18. Check whether a person is eligible to vote

```
#!/bin/bash
echo "Enter your age:"
read age
if [ "$age" -ge 18 ]
then
    echo "You are eligible to vote"
else
    echo "You are not eligible to vote"
fi
```